EXP NO: 20 EXPERIMENT TO UNDERSTAND PANDAS LIBRARY USE CASES IN DATA SCIENCE

AIM:

Understand data frame use cases using pandas library.

PYTHON CODE:

```
import numpy as np
import pandas as pd
list = [[1, 'Smith', 50000],
        [2, 'Jones', 60000]]

df = pd.DataFrame(list)
print(df)
df.columns = ['Empd', 'Name', 'Salary']
print(df)
df.info()
df = pd.read_csv("/content/50_Startups.csv")
df.info()
print(df)
print(df.head())
print(df.tail())
df = pd.read_csv("/content/employee.csv")
print(df.head())
print(df.tail())
df.info()
print(df.salary)
print(type(df.salary))
print(df.salary.mean())
print(df.salary.median())
print(df.salary.mode())
print(df.salary.var())
print(df.salary.std())
print(df.describe())
print(df.describe(include='all'))
empCol = df.columns
print(empCol)
emparray = df.values
print(emparray)
employee_DF = pd.DataFrame(emparray, columns=empCol)
print(employee_DF)
```

OUTPUT:

```
   0      1      2
0  1  Smith  50000
1  2  Jones  60000
   Empd   Name  Salary
0     1  Smith   50000
1     2  Jones   60000
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Empd    2 non-null      int64
 1   Name    2 non-null      object
 2   Salary  2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   R&D Spend        5 non-null      float64
 1   Administration   5 non-null      float64
 2   Marketing Spend  5 non-null      float64
 3   State            5 non-null      object
 4   Profit           5 non-null      float64
dtypes: float64(4), object(1)
memory usage: 332.0+ bytes
   R&D Spend  Administration  Marketing Spend       State     Profit
0  165349.20       136897.80        471784.10    New York  192261.83
1  162597.70       151377.59        443898.53  California  191792.06
2  153441.51       101145.55        407934.54     Florida  191050.39
3  144372.41       118671.85        383199.62    New York  182901.99
4  142107.34        91391.77        366168.42     Florida  166187.94
   R&D Spend  Administration  Marketing Spend       State     Profit
0  165349.20       136897.80        471784.10    New York  192261.83
1  162597.70       151377.59        443898.53  California  191792.06
2  153441.51       101145.55        407934.54     Florida  191050.39
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3  144372.41       118671.85        383199.62    New York  182901.99
4  142107.34        91391.77        366168.42     Florida  166187.94
```

```
   emp id                 name  salary
0       1     SREE VARSSINI K S    5000
1       2           SREEMATHI B    6000
2       3               SREYA G    7000
3       4  SREYASKARI MULLAPUDI    5000
4       5         SRI AKASH U G    8000
   emp id                  name  salary
2       3                SREYA G    7000
3       4   SREYASKARI MULLAPUDI    5000
4       5          SRI AKASH U G    8000
5       6  SRI HARSHAVARDHANAN R    3000
6       7  SRI HARSHAVARDHANAN R    6000
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   emp id  7 non-null      int64
 1   name    7 non-null      object
 2   salary  7 non-null      int64
dtypes: int64(2), object(1)
memory usage: 300.0+ bytes
0    5000
1    6000
2    7000
3    5000
4    8000
5    3000
6    6000
Name: salary, dtype: int64
<class 'pandas.core.series.Series'>
5714.285714285715
6000.0
0    5000
1    6000
Name: salary, dtype: int64
2571428.5714285714
1603.5674514745463
         emp id       salary
count  7.000000     7.000000
mean   4.000000  5714.285714
std    2.160247  1603.567451
min    1.000000  3000.000000
25%    2.500000  5000.000000
50%    4.000000  6000.000000
75%    5.500000  6500.000000
max    7.000000  8000.000000
```

```
          emp id                    name       salary
count   7.000000                       7     7.000000
unique       NaN                       6          NaN
top          NaN  SRI HARSHAVARDHANAN R          NaN
freq         NaN                       2          NaN
mean    4.000000                     NaN  5714.285714
std     2.160247                     NaN  1603.567451
min     1.000000                     NaN  3000.000000
25%     2.500000                     NaN  5000.000000
50%     4.000000                     NaN  6000.000000
75%     5.500000                     NaN  6500.000000
max     7.000000                     NaN  8000.000000
Index(['emp id', 'name', 'salary'], dtype='object')
[[1 'SREE VARSSINI K S' 5000]
 [2 'SREEMATHI B' 6000]
 [3 'SREYA G' 7000]
 [4 'SREYASKARI MULLAPUDI' 5000]
 [5 'SRI AKASH U G' 8000]
 [6 'SRI HARSHAVARDHANAN R' 3000]
 [7 'SRI HARSHAVARDHANAN R' 6000]]
  emp id                  name salary
0      1      SREE VARSSINI K S   5000
1      2            SREEMATHI B   6000
2      3               SREYA G   7000
3      4   SREYASKARI MULLAPUDI   5000
4      5          SRI AKASH U G   8000
5      6  SRI HARSHAVARDHANAN R   3000
6      7  SRI HARSHAVARDHANAN R   6000
```

RESULT:

Thus the code is executed successfully.